A. (a) After playing around with g and different choices of w and y, I came to the conclusion that g must be a scalar multiple of the the identity matrix $\mathbb{1}$.

(b) *Proof.* Let g be a scalar multiple of the identity matrix. Let w, y be vectors in $\mathbb{R}^2$. Then their dot product is $\frac{w^T \mathbb{1} y}{\sqrt{w^T \mathbb{1} w}\sqrt{y^T \mathbb{1} y}}$. The angle between two vectors pushed forward by dx_q is $\frac{w^T g y}{\sqrt{w^T g w}\sqrt{y^T g y}}$. When $g = c\mathbb{1}$, this simplifies to $\frac{cw^T \mathbb{1} y}{\sqrt{cw^T \mathbb{1} w}\sqrt{cy^T \mathbb{1} y}} = \frac{w^T \mathbb{1} y}{\sqrt{w^T \mathbb{1} w}\sqrt{y^T \mathbb{1} y}}$. So, dx_q is angle preserving. □

B. (a) The g associated with $x(u, v) = (\frac{2u}{u^2+v^2+1}, \frac{2v}{u^2+v^2+1}, \frac{u^2+v^2-1}{u^2+v^2+1})$ is $g_x = \frac{4}{(u^2+v^2+1)^2}\mathbb{1}$. No matter what q is selected, g_x is always a scalar multiple of the identity, and thus preserves angles.

(b) When $q = (0, 0)$, g_x is $4\mathbb{1}$, and thus $\sqrt{\det g} = 4$ and dx_q does not preserve area at this point. When q is a point on the unit circle, $u^2 + v^2 = 1$, and $g_x = \mathbb{1}$, whose determinant is 1. Therefore, on the unit circle, dx_q preserves area. As q gets arbitrarily far away from the origin, g approaches the 0 matrix, and thus the determinant is 0, which does not preserve area.

C. Let $U = (-\pi, \pi) \times (-1, 1)$. We can send U to T by $r : U \to T$ given by $r(u_1, u_2) = (\cos u_1, \sin u_1, u_2)$. To project this onto the unit sphere, we need to make sure that $|r(u, v)|^2$ is 1, but we also must keep the height the same, thus fixing r_3. So, we compute $|r(u, v)|^2 = \cos^2 u + \sin^2 u + v^2$. To make this 1, we can have $x(u, v) = (\sqrt{1 - v^2}\cos u, \sqrt{1 - v^2}\sin u, v)$. This is a local parameterization of $\mathbb{S}^2$. We now show that it is area preserving. We compute that $\frac{\partial x}{\partial u} = (-\sqrt{1 - v^2}\sin u, \sqrt{1 - v^2}\cos u, 0)$. We also compute that $\frac{\partial x}{\partial v} = (\frac{-v\cos u}{\sqrt{1-v^2}}, \frac{-v\sin u}{\sqrt{1-v^2}}, 1)$. We compute that $g = \begin{bmatrix} 1 - v^2 & 0 \\ 0 & \frac{1}{1-v^2} \end{bmatrix}$. We can see that $\det g = 1$, and thus $\sqrt{\det g} = 1$, and x is area preserving.